

LC 2696

Minimum String length after removing substrings

IBM

JPMC

given a string S, in one ops one can remove any occurrence of substring "AB" or "CD" from S

Key Insight

string concatenates after removing the substring and could produce new "AB" or "CD" substrings

↳ *the biggest hint for using stacks*

Logic Flow

so basically whenever '**B**' comes we see

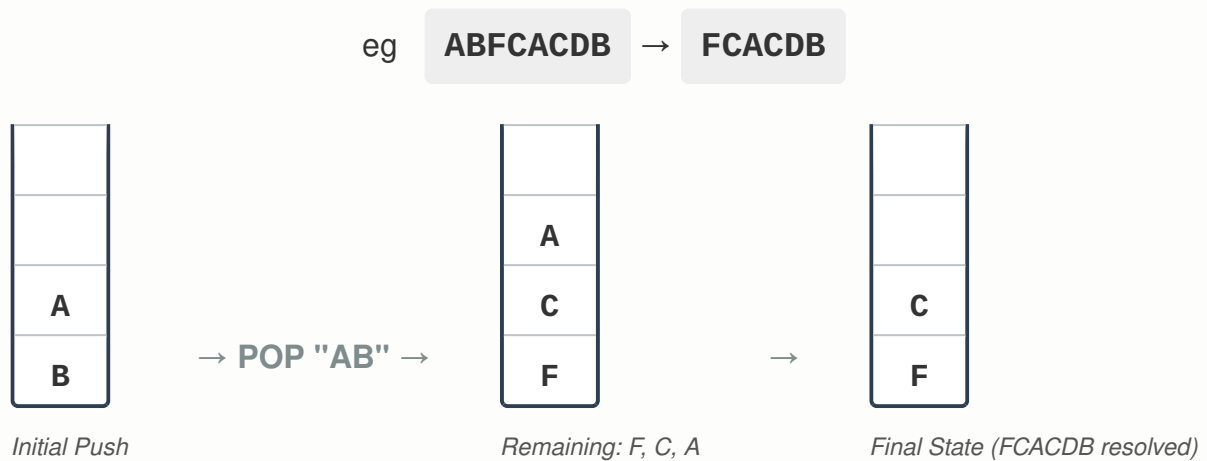
if top is '**A**'

→ yes → **POP**

→ no → **push**

similarly for "CD"

Example Trace



C++ Implementation

```
class Solution { public: int minLength(string s) { stack<char> st; for(char c : s){ if(c == 'B' && !st.empty() && st.top() == 'A'){ st.pop(); } else if(c == 'D' && !st.empty() && st.top() == 'C'){ st.pop(); } else{ st.push(c); } } return st.size(); } };
```